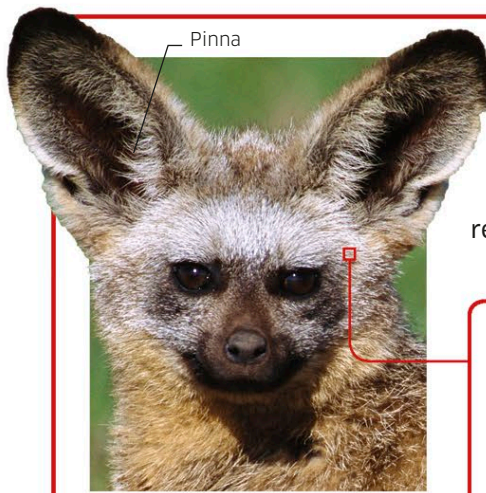


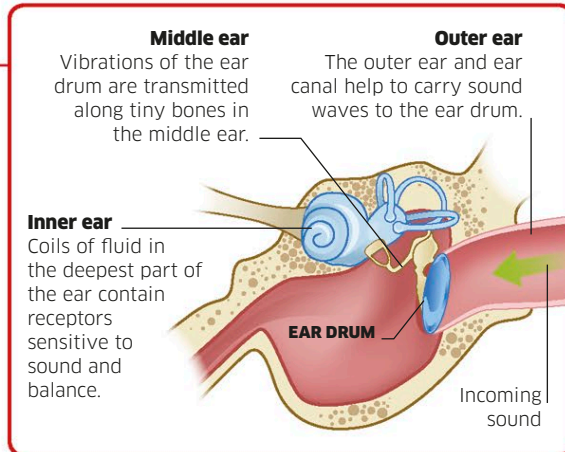


Bat-eared fox

In mammals, each ear opening is surrounded by a fleshy funnel for collecting sound, called a pinna. The desert-living bat-eared fox has such large pinnae that it also uses them to radiate warmth to stop it from overheating.

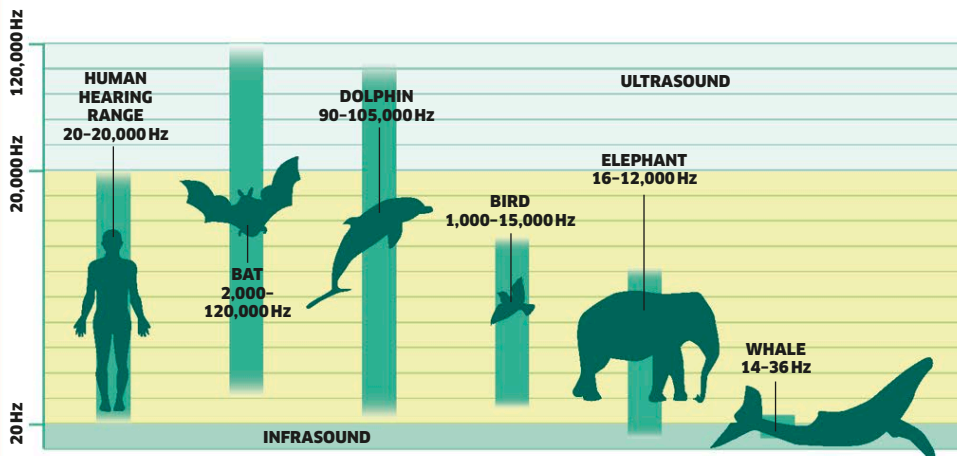
Hearing

Animals hear because their ears contain receptors that are sensitive to sound waves. As the waves enter the ear, they vibrate a membrane called an ear drum. The vibrations pass along a chain of tiny bones until they reach the receptors within the inner ear.



Hearing ranges

The pitch, or frequency, of a sound is measured in hertz (Hz): the number of vibrations per second. Different kinds of animals detect different ranges of pitch, and many are sensitive to ultrasound and infrasound that are beyond the hearing of humans.



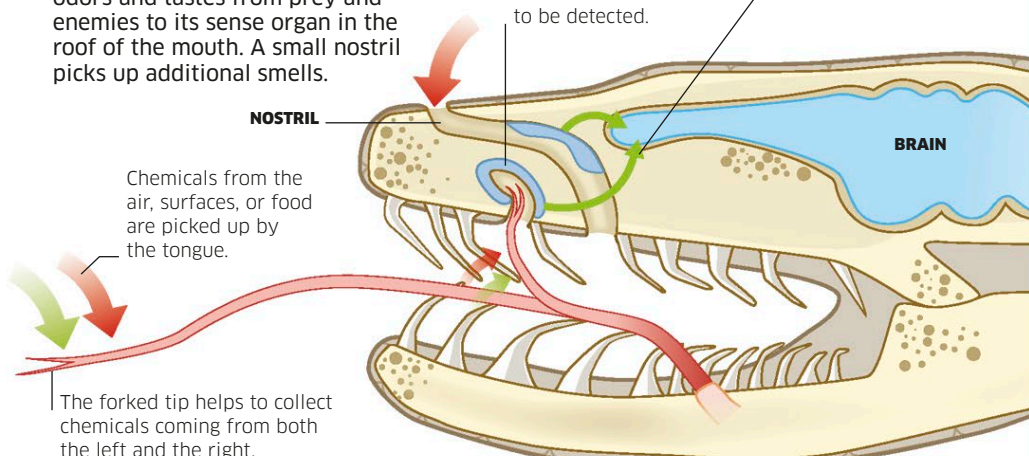
Snake senses

A snake's tongue has no receptors and instead is used for transferring odors and tastes from prey and enemies to its sense organ in the roof of the mouth. A small nostril picks up additional smells.

Jacobson's organ

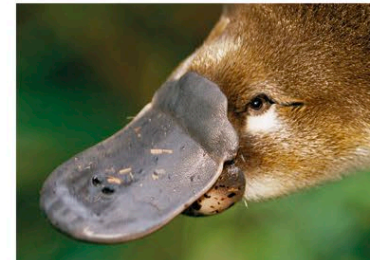
Chemicals on the tongue tip are transferred here to be detected.

Smell receptors pass signals along nerves to the brain.



Other ways of sensing the world

The lives of many kinds of animals rely on quite extraordinary sensory systems. Some have peculiar types of receptors that are not found in other animals. These give them the power to sense their surroundings in ways that seem quite unfamiliar to us—such as by picking up electrical or magnetic fields.



Electroreception

The rubbery bill of the platypus—an aquatic egg-laying mammal—contains receptors that detect electrical signals coming from the muscles of moving prey. They help the platypus find worms and crayfish in murky river waters.



Echolocation

Bats and dolphins use echolocation to navigate and find food. By calling out and listening for the echoes bouncing back from nearby objects, they can work out the positions of obstacles and prey.



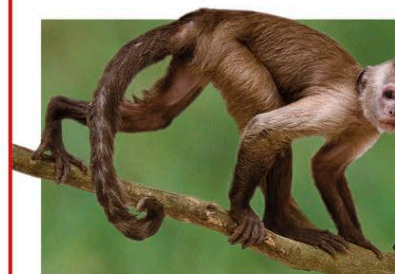
Fire detection

Most animals flee from fire, but the fire beetle thrives near flames. Its receptors pick up the infrared radiation coming from a blaze, drawing it to burned-out trees where it can breed undisturbed by predators.



Magnetoreception

Birds can sense the earth's magnetic field. By combining this with information about the time of day and position of the sun or stars, they can navigate their way on long-distance migrations.



Balance

All vertebrate animals have balance receptors in their ears to sense the position of their head and tell up from down. These help humans walk upright and stop climbing animals, such as capuchin monkeys, from falling out of trees.



Time

Tiny animals, such as insects, experience time more slowly because their senses can process more information every second. Compared with humans, houseflies see everything in slow motion—helping them to dodge predators.